Operating-System (OS) Project Report

Text Editor with Database Integration and Security

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1. Abstract

This project presents a professional desktop-based text editor developed in Java. The application offers a user-friendly interface for creating, editing, and managing text documents. It incorporates advanced features such as **undo/redo functionality, customizable fonts and colors, search and highlight capabilities, and secure storage in a MySQL database with optional password protection**. Users can save and open files from both **local disk** and **database**, ensuring flexibility, data persistence, and security. The project demonstrates practical application of **object-oriented programming, GUI development, event-driven programming, and database management**.

2. Introduction

Text editors are essential tools for academic, professional, and personal use. This project focuses on developing a **feature-rich Java text editor** that enhances the traditional editing experience with advanced functionalities and secure storage. The system combines **graphical user interface design, file handling, and database integration** to offer a seamless and secure text editing environment.

3. Objectives

- Develop a professional text editor with an intuitive GUI using Java Swing.
- Implement standard file operations: New, Open, Save, Save As, and Exit.
- Enable text formatting: font selection, font size adjustment, and color customization.
- Provide editing functionalities: undo, redo, copy, cut, paste, delete, select all, and search with highlighting.
- Support secure storage of files in a MySQL database with optional password protection.
- Allow users to manage files from both **disk** and **database**.
- Ensure a responsive, stable, and user-friendly application.

4. Tools and Technologies Used

Component	Technology / Version
Programming Language	Java SE 8+

GUI Framework	Swing/AWT
Database	MySQL 8+
IDE	IntelliJ IDEA / Eclipse
Build / Run	JDK 8+, JDBC Connector

5. System Architecture

The system follows a modular architecture to separate concerns and maintain scalability:

- **GUI Layer:** Provides interactive components for editing and formatting text.
- **File Management Layer:** Handles local disk operations.
- **Database Layer:** Manages secure storage and retrieval of files in MySQL.
- **Undo/Redo Management:** Tracks changes and allows reverting actions.
- **Search and Highlight Functionality:** Enables searching for specific text and highlights matches for easy identification.

6. Key Features

1. **File Operations:** New, Open (disk/database), Save (disk/database), Exit with save prompt.
2. **Text Editing:** Undo, Redo, Copy, Cut, Paste, Delete, Select All.
3. **Text Formatting:** Font type selection, font size adjustment, text color customization.
4. **Search Functionality:** Search for words and highlight occurrences.
5. **Database Integration:** Secure storage and retrieval of files with optional password protection.

7. Database Design

Database Name: textEditorDB

Table: files

Column	Data Type	Description
id	INT	Primary key
name	VARCHAR(255)	File name
content	LONGTEXT	File content
password	VARCHAR(128)	Optional password for security
created_at	TIMESTAMP	Creation timestamp
modified_at	TIMESTAMP	Last modification timestamp

8. User Interface Design

- **Main Editor Window:** Central text area with vertical scroll bar; resizable interface.
- **Menu Bar:** File, Edit, and Format menus for all operations.
- **Dialogs:** File chooser dialogs, password prompts, search input, and notifications for user actions.
- **Responsiveness:** The interface adapts to window resizing for better usability.

9. Testing and Validation

- Verified all menu operations, editing, and formatting features.
- Tested saving and opening files on both disk and database, including password-protected files.
- Handled edge cases such as empty files, cancelled dialogs, and incorrect passwords.
- Confirmed all functionalities operate correctly without errors or crashes.

10. Challenges Faced

- Implementing password-protected database storage and verification.
- Maintaining consistency in undo/redo functionality for key binding.
- Search and Highlight Function.
- Integrating file system and database operations seamlessly.

11. Conclusion

The Advanced Java Text Editor successfully demonstrates the development of a professional desktop application combining **text editing, formatting, search functionality, and secure database storage**. The project highlights practical applications of **GUI development, event-driven programming, and database integration**, producing a robust, secure, and user-friendly text editor suitable for both academic and professional use.

12. Future Enhancements

- Implement **auto-save and version control** for database files.
- Encrypt database passwords for enhanced security like hashing.
- Add **syntax highlighting** for programming languages.
- Integrate **cloud storage options** for remote access.
- Enable **multi-tab editing** to work on multiple documents simultaneously.

13. GitHub Repository

GitHub Repository Link: https://github.com/raheelnaziir/Text-Editor.git